

Michael J Lalonde

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Engineering leader and active technical contributor. Over two decades building and scaling high-volume, low-latency systems and the teams behind them. Continuously seek to improve through new challenges and perspectives. Driven by owning architecture, shipping production code, and growing engineering talent.

Experience

MachineQ, a Comcast company

Philadelphia, PA

Director, Product Engineering

2018–Present

- Designed and implemented platform optimizations that cut cloud costs by 80%, reduced data processing from seconds to milliseconds, and scaled throughput 100x beyond original capacity.
- Established code quality standards and peer review processes that cut production incidents by >95%, from monthly occurrences to nearly eliminated.
- Developed backward compatibility protocols enabling confident weekly deployments that accelerated feature delivery with zero customer disruptions.
- Transformed engineering culture through strategic focus on ownership and technical excellence, resulting in promotions for 50% of team members and conversion of contractors to full-time roles.

Comcast

Philadelphia, PA

Engineering Manager

2014–2018

- Directed team of Java engineers building mission-critical systems with 99.99% uptime for infrastructure serving 25+ million Comcast customers.
- Championed DevOps transformation that boosted production stability by 60% and slashed deployment times by 50%, enabling near daily releases.
- Fostered a culture of technical exploration, encouraging the team to evaluate and adopt technology stacks (Golang, Kafka, etc.) that better fit problem domains, resolving scaling bottlenecks and reducing infrastructure costs by 40%.

Comcast

Philadelphia, PA

Senior Software Engineer

2008–2014

- Designed the architecture and delivered multiple high-performance platforms using Java, Golang, Erlang, and Clojure.
- Optimized service architecture to handle 10x traffic growth while reducing hardware footprint by 25%.
- Led adoption of continuous integration practices and automated testing that reduced regression bugs by 70%.
- Developed systematic troubleshooting methodologies that reduced mean time to resolution by 65%.

Technical Skills

Languages: Go, SQL, JavaScript, C, Python, HTML/CSS

Infrastructure: Docker, Kubernetes, Terraform, Azure Cloud, Hashicorp Vault, Nginx, Apache

Data Systems: PostgreSQL, Redis, Kafka, SQLite

Tools & Observability: Git, Grafana, KQL, Make, Emacs, Node

Projects

Animated Brewing

Coatesville, PA

Automated Keg Washing System

2020

Designed and built an automated keg washing system for a local craft brewery, cutting cleaning time in half and manual labor by 80%.

- Programmed a Teensy microcontroller in C to control automated valve sequences and cleaning cycles.
- Engineered complete system architecture including piping diagrams, electrical schematics, and component selection.
- Delivered solution at 15% of commercial system cost while maintaining industry cleaning standards.

Ethereum Network

Home

Ethereum Validator Node

2019

Built and deployed an Ethereum validator node for the Beacon Chain, achieving 99.8% uptime and consistent attestation performance.

- Configured optimized hardware and networking setup for 24/7 reliable operation.
- Developed custom monitoring solution using Telegram bot for real-time alerts and performance tracking.

Earlier Experience

SunGARD

Philadelphia, PA

Software Engineer

2003–2008

- Built network monitoring systems processing millions of events daily with custom correlation engine that reduced false-positive alerts by 85%.
- Designed and managed data warehouse solution with automated rollups and aggregations for long-term metrics storage.

Cadient Group

Conshohocken, PA

IT Engineer

2002–2003

- Led infrastructure planning and implementation for growing startup (domain, email, web, backup), 10% under budget with zero-disruption migration.

Qwest Global Services

Philadelphia, PA

Solutions Engineer

2000–2002

- Technical lead on enterprise client projects delivering custom applications and solutions.
- Developed document automation system for enterprise workflows.
- Built Palm Pilot newspaper application demonstrating early mobile development expertise.

IKON Office Solutions

Malvern, PA

Web Developer

1999–2000

- Led design and implementation of corporate intranet serving 40,000+ employees globally.
- Created custom web applications and content search solutions for company divisions.